

## Definition

**Insertion Sort** is a simple sorting algorithm that builds the sorted array one element at a time. It takes each element and inserts it into its correct position among the previously sorted elements.

## Algorithm

1. Start from the second element of the array.
2. Store the current element in key.
3. Compare key with the elements before it.
4. Shift elements greater than key one position to the right.
5. Insert key into its correct position.
6. Repeat until all elements are sorted.

Output:

```
Sorted array: [5, 6, 11, 12, 13]
...Program finished with exit code 0
Press ENTER to exit console.□
```

## Time Complexity

| Case         | Complexity |
|--------------|------------|
| Best Case    | $O(n)$     |
| Average Case | $O(n^2)$   |
| Worst Case   | $O(n^2)$   |

- **Best case:** Array is already sorted.
- **Worst case:** Array is sorted in reverse order.

## Space Complexity

**$O(1)$**  — Insertion Sort sorts the array **in-place** and uses only a constant amount of extra memory.